



OPEN LINC: a framework for maintaining high-quality passive data in digital phenotyping studies

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Achieving high-quality passive data in smartphone-based digital phenotyping studies remains a significant challenge in real-world settings where technical barriers and inconsistent user engagement limit continuous data collection. In response to the field's overreliance on imputation methods and model-based approaches, we present LINC, a framework of best practices that supports the collection of high-quality passive data in digital phenotyping research. LINC, which stands for Launch, Interact, Notify, and Correct, systematizes operational procedures across four domains: (a) device and app configuration, (b) participant engagement with the app, (c) real-time data monitoring, and (d) troubleshooting disruptions in passive data collection. Each component is accompanied by practical resources designed to facilitate implementation without requiring specialized technical expertise. To demonstrate feasibility, we applied the LINC framework to a large observational study ($n = 373$) examining social media use and youth mental health over a two-to three-week period. Our study, following this systematized approach, achieved a median GPS-based passive data quality of 0.92 (IQR: 0.59–0.98), with 75% of participants surpassing 0.59 and 25% exceeding 0.98. These results compare favorably to published benchmarks and demonstrate that it is feasible to achieve high-quality passive data in digital phenotyping research.

Keywords Digital phenotyping, Data quality, Missingness, Passive sensing, Framework, Smartphone sensors

Over the past decade, smartphone-based technologies have enabled researchers to better understand mental illness beyond the clinic. Central to this progress is digital phenotyping, the real-time, in situ quantification of behavior using smartphone sensor data (e.g., GPS, accelerometer)¹. This approach enables the unobtrusive collection of large volumes of behavioral data from individuals with mental illness, with the potential to reach billions of smartphone users globally². Research leveraging these passive data streams has shown promise in characterizing symptoms³, monitoring symptom trajectories⁴, predicting relapse⁵, and supporting personalized interventions⁶. Despite these advances, ensuring high-quality passive data with minimal missingness remains a major challenge for generating reliable and reproducible findings.

This limitation is well recognized in the literature. Recent systematic reviews have underscored the high prevalence of missing sensor data and the inconsistent reporting of passive data quality across digital phenotyping studies^{7,8}. A systematic review of 203 studies using digital phenotyping to monitor psychosis and schizophrenia⁹ reported missingness rates ranging from 19% to 72%. Similarly, Kiang and colleagues¹⁰ reviewed six studies comprising 211 participants and found that baseline missingness was as high as 27%. Beyond overall prevalence estimates, evidence also highlights the rapid onset of passive data degradation. Currey and Torous¹¹ reported that data quality from GPS and accelerometer sensors can decline precipitously within just three days if participants do not open the digital phenotyping app. Collectively, these results indicate that poor passive data quality (i.e., high missingness) is not an isolated occurrence in a subset of studies or a particular app but rather a systemic challenge for the field that requires transparent reporting and proactive mitigation to enable the generation of reliable and reproducible digital phenotyping research. Prior work in the field has identified several key contributors to low passive data quality and missingness, including (1) hardware malfunction¹², (2) improper sensor or application configuration¹¹, (3) operating system (OS) or permission-related restrictions^{13,14}, and (4) user disengagement (e.g., not regularly opening the app)¹¹.

The consequences of poor passive data quality in digital phenotyping studies are both methodological and clinical, threatening the interpretability of downstream findings. Missing or inconsistent sensor data can bias derived social and behavioral features, distort associations with clinical outcomes, and reduce statistical power.

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For example, Currey and Torous¹¹ demonstrated that increasing levels of missingness were associated with greater error in GPS-derived measures such as home time and entropy. Inaccurate or missing GPS data can also lead to erroneous inferences about mobility patterns, such as misclassifying time spent at home as time spent at the gym. Similarly, Shen and colleagues¹⁵ reported that GPS-derived home time and accelerometer-based features used to predict functional status performed poorly at higher levels of missingness. This issue is widespread as noted by Sameh and colleagues⁸ in their systematic review of 66 machine learning-based digital phenotyping studies, who reported that low sampling rates and unaddressed missing data were significant challenges identified across the reviewed studies.

In response to this ongoing challenge, researchers have employed a wide range of strategies to improve passive data quality in digital phenotyping studies, spanning design-based, statistical imputation, and model-based approaches. One widely used design strategy involves implementing planned sensor sampling schedules, in which data are collected according to predefined on-off cycles (e.g., 2 min of GPS collection followed by 10 min of rest). This approach reduces battery drain and, in turn, mitigates operating system-level restrictions on background data collection. Importantly, these structured gaps allow researchers to distinguish expected missingness arising from the sampling protocol from unexpected data loss, thereby enabling more accurate estimation of sensor coverage and the application of tailored imputation strategies. Leveraging this design, Barnett and Onnela¹⁶ imputed missing mobility data using a weighted statistical resampling approach, achieving significantly lower interpolation error than standard linear methods.

Beyond design-based strategies, prior work has largely relied on post hoc methods to address missing data. In a recent systematic review, Bladon and colleagues⁹ reported that the most common approaches involved simple imputation techniques, such as replacing missing values with the mean over the recording period or the nearest observed value. More advanced methods included regularized iterative principal components analysis and multiple imputation by chained equations (MICE), which estimate missing values using predictive models derived from observed data. Other approaches have modeled missingness directly, for example, one study used logistic regression to model predictors of missingness using participant baseline characteristics (e.g., demographics, symptom severity) and incorporated these variables into downstream intention-to-treat (ITT) mixed-effects models¹⁷. Alternatively, some studies adopted data exclusion strategies, omitting days with insufficient sensor coverage (e.g., fewer than 50 data points), often set as an arbitrary threshold from predictive modeling^{18,19}.

While imputation and model-based approaches may be effective in salvaging incomplete passive data streams, they risk obscuring clinically valuable signals by reducing the idiosyncratic nature of patient-level data¹⁵. Even when applied appropriately, these methods have the potential to inadvertently bias derived behavioral features and, in turn, compromise the validity of downstream analyses. Although much of the field's current efforts are focused on post hoc correction of already missing data^{9,20,21}, we argue that greater emphasis should be placed on prevention of missing data rather than relying on modeled approximations. In our previous work¹¹, we have suggested preventive strategies that may be helpful to reduce missingness in digital phenotyping studies, such as implementing real-time data monitoring dashboards and automated reminders to prevent participant disengagement. Building upon these efforts and our experience implementing digital phenotyping protocols^{22,23}, we systematized our operational practices into the LINC (Launch, Interact, Notify, and Correct) framework, which we applied to a two-to three-week study evaluating the effect of social media use on youth mental health using digital phenotyping²⁴. Thus, this framework provides tools and strategies for maintaining high-quality passive data in smartphone-based digital phenotyping studies by addressing the core challenges in passive data collection.

While individual components of the LINC framework have been described in prior digital phenotyping studies^{11,25,26}, no prior work has integrated these elements into a cohesive, operationalized framework with accompanying implementation tools. LINC's contribution lies not in introducing novel techniques, but in systematizing established best practices into a reproducible workflow supported by practical resources, including configuration checklists, monitoring scripts, troubleshooting templates, and participant outreach protocols. By lowering the technical and operational barriers to adoption, this systematization enables research teams to implement effective data quality practices without extensive trial-and-error or specialized infrastructure, addressing a critical gap in the field's ability to achieve high-quality passive data collection.

Thus, this manuscript has three primary goals: (1) to present LINC as a framework that systematizes best practices into a standardized workflow, with practical implementation resources for each of its components; (2) to demonstrate the feasibility of this approach through findings from our social media study; and (3) to illustrate the importance of passive data quality for accurate feature derivation and valid downstream analyses.

Methods

This section outlines the LINC framework applied to maintain high passive data quality in our large digital phenotyping study ($n=373$ analyzed participants) examining the relationship between social media use and youth mental health²⁴. We begin by briefly describing the procedures of the study, followed by a detailed explanation of how passive data were collected and data quality was calculated (Fig. 1). Finally, we present the key components of the LINC Framework (Fig. 2) applied in this study, which included: (a) configuring and maintaining optimal device and app-level settings, (b) encouraging sustained participant app engagement, (c) real-time data monitoring procedures, and (d) identifying and troubleshooting disruptions in passive data collection.

Social media study overview

This observational digital phenotyping study recruited young adults aged 18 to 24 years, primarily college students, with minimal baseline psychiatric symptoms (non-clinical sample). Participants first completed a two-

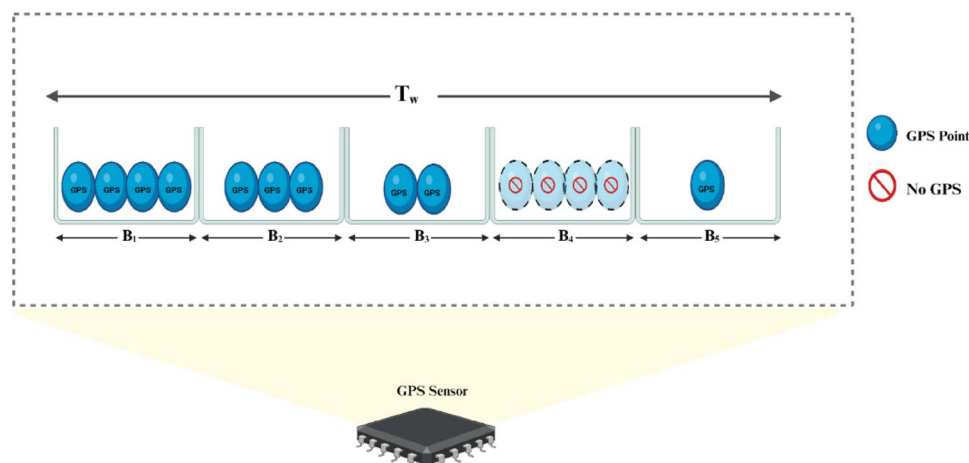


Fig. 1. Method for calculating passive data quality using GPS binning. This figure illustrates how passive data quality was calculated using GPS data. Each 24-hour day (T_w) was divided into 144 non-overlapping 10-minute bins ($B_1, B_2, \dots, B_{144}$). A bin was considered “covered” (blue circle) if at least one GPS data point was recorded during that interval, and “missing” (red strike) if no data were present. Passive data quality was defined as the proportion of bins with at least one GPS reading.

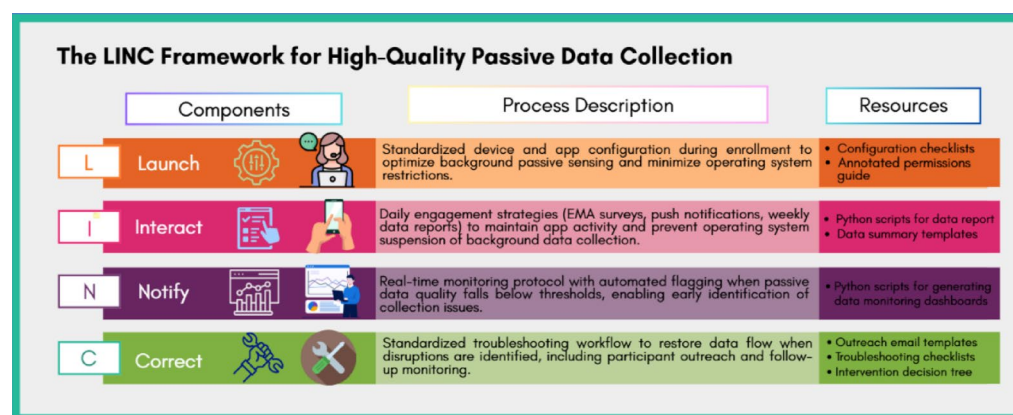


Fig. 2. The LINC (Launch, Interact, Notify, Correct) Framework for maintaining high-quality passive data collection in digital phenotyping studies. Each component includes a process description and associated resources available in the Supplementary Materials.

week observational phase involving passive smartphone sensing, weekly standardized mental health surveys, and daily ecological momentary assessments (EMAs). This was followed by an optional one-week social media detox intervention, in which participants were asked to abstain from using five major social media platforms: Facebook, Instagram, Snapchat, TikTok, and X (formerly Twitter). Participants completed two virtual study visits over two weeks, with an additional visit for participants completing the detox. At Visit 1, they provided electronic informed consent, completed baseline assessments, and configured the mindLAMP digital phenotyping app on their smartphones. During the two-week observational phase, daily passive sensing data (e.g., GPS, accelerometer) and daily EMAs were collected. At Visit 2, participants repeated clinical assessments (excluding demographics), reviewed their personalized data summary (Fig. 4) with a research assistant, and reported their social media app-specific usage metrics. Participants then chose whether to complete a one-week social media detox while continuing passive sensing and EMAs. At Visit 3, they completed final assessments, reviewed their detox data report (Fig. 4), and submitted app-specific usage metrics for the detox period. A detailed description of the study procedure is available in the Supplementary Methods.

Ethical considerations

The social media study was approved by the Institutional Review Board (IRB) at Beth Israel Deaconess Medical Center (protocol #2023P001009), and methods were conducted in accordance with institutional guidelines. Informed consent was obtained and documented from all participants via REDCap (version 14.0.30). Data were stored in HIPAA-compliant systems and de-identified prior to conducting any analyses.

Study passive data collection

Passive data were captured by the mindLAMP app²⁷ and included GPS, accelerometer, and screen state. The app sampled GPS data at a rate of 1 Hz, meaning it requested location updates every second from the phone’s location services. Once collected, the open-source Python library Cortex²⁸ was used to calculate GPS-based passive data quality and related mobility features derived from the data. GPS data were used for passive data calculations in this paper as they are less computationally expensive and faster to process relative to accelerometer data. In addition, we have demonstrated that GPS and accelerometer data quality are correlated, validating the use of GPS as a reliable proxy for overall passive data quality (Supplementary Fig. S3). In this study, passive data quality was calculated daily for each participant and operationalized as the proportion of 10-minute intervals within a 24-hour period that contained at least one GPS data point. A standard 24-hour day is divided into 144 non-overlapping 10-minute bins for the purpose of calculating passive data quality. As illustrated in Fig. 1, a typical 24-hour time window (T_w) is divided into 10-minute bins, denoted $B_1, B_2, \dots, B_{144}$. A bin is marked as “covered” if it includes at least one GPS ping (blue circle), and “missing” if no GPS data are present (red strike). In the Fig. 1 example, 4 out of 5 non-overlapping 10-minute bins contain valid GPS data, resulting in a passive data quality score of 0.80 for the illustrated time window, T_w . This approach provides an interpretable way for assessing daily passive sensing coverage from smartphones in digital phenotyping studies.

LINC framework to maintain high data quality

The LINC Framework (Launch, Interact, Notify, and Correct) systematizes operational procedures for maintaining high passive data quality in digital phenotyping studies. The framework is built around four central domains, each targeting key sources of missingness: (a) *Launch*: configuring and maintaining optimal device and app-level settings, (b) *Interact*: encouraging sustained participant app engagement, (c) *Notify*: real-time data monitoring procedures, and (d) *Correct*: identifying and troubleshooting disruptions in passive data collection. Each component is supported by practical resources that can be readily adopted by other research teams. These materials are included in the Supplementary Materials online and are carefully designed to support the reproducible deployment of high-quality passive data collection protocols.

Launch: configuring and maintaining optimal device and app-level settings

For consistent passive data collection, a standardized onboarding protocol was applied across all participant devices during study enrollment, led by a research assistant (RA). This included both device-level and app-level (mindLAMP) configurations, operationalized as a toolkit of checklists and annotated screenshots of the process intended to optimize background passive sensing and minimize interruptions due to operating system restrictions. At the device level, participants were instructed by staff during onboarding to set location permissions to “Always Allow” (iOS) or “Allow all the time” (Android). Battery optimization settings (e.g., low power mode, adaptive battery) were also reviewed and adjusted to prevent mindLAMP from being suspended when not actively in use (Fig. 3). To enable full passive sensing functionality, participants were asked to install and configure Apple Health (iOS) or Google Fit (Android), which serve as intermediaries that augment the

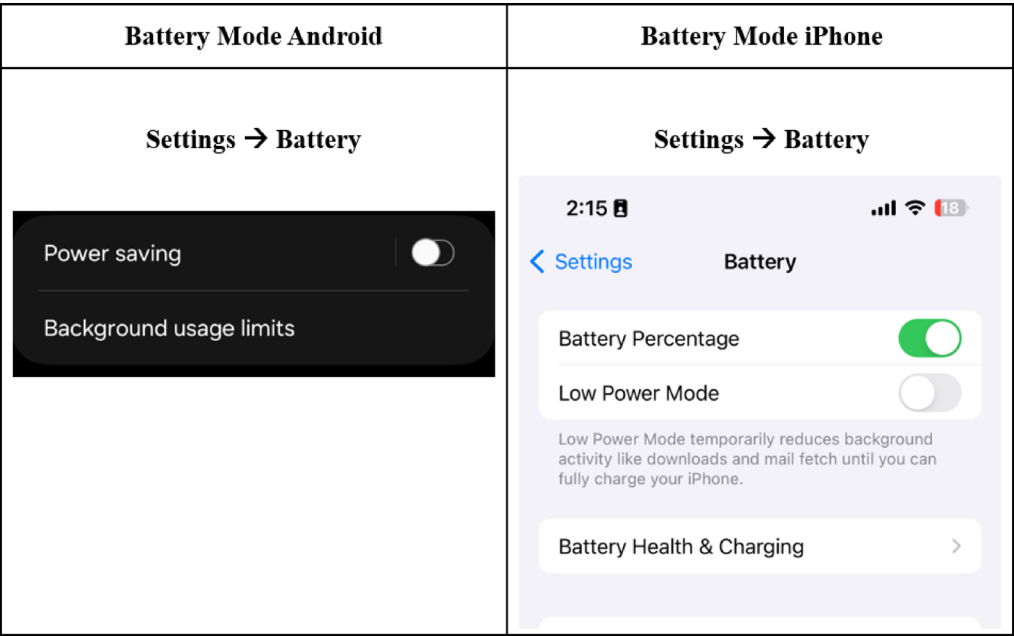


Fig. 3. Battery optimization settings for Android and iPhone devices used during onboarding. Participants were guided to disable Power Saving Mode and remove Background Usage Limits on Android (left), and to ensure Low Power Mode was turned off on iPhone (right), as both settings can interfere with continuous passive data collection in the mindLAMP app.

collection of sensor data necessary for deriving features such as step count. At the app level, the sampling rates for GPS (1 Hz) and accelerometer (5 Hz) data were configured on each participant's device.

To standardize the onboarding workflow and reduce participant to participant variability, RAs used a structured step-by-step checklist outlining the key configuration steps supporting uninterrupted data collection (see Supplementary Materials, “Onboarding Configuration Checklist”). After the onboarding session, a checklist with annotated screenshots of the configuration settings was emailed to each participant (see Supplementary Materials, “Participant Annotated Permissions Configuration”). At each point of contact, research staff advised participants to avoid letting their device enter low power mode by keeping their phones charged, to avoid disabling location services, and to refrain from force quitting the app. These behavioral reinforcements were intended to complement the technical setup and ensure continuous passive data collection by the mindLAMP app over the course of the study. All materials used, including the onboarding configuration checklists and OS permissions guide are provided in Supplementary Materials online and can be adapted by other digital health research teams.

Interact: app-based strategies for participant engagement

Sustained engagement with the mindLAMP app was important for maintaining uninterrupted passive data collection. Modern smartphone operating systems (iOS 15+, Android 12+) implement aggressive background process management that restricts sensor access for apps not regularly used. By requiring daily app interaction, we aimed to maintain the app in active status, preventing operating system suspension of background data collection. In the social media study, we applied a structured engagement protocol during onboarding, reinforced through automated reminders and weekly participant data sharing. A key component of this strategy was to include daily EMA surveys for participants to complete that assessed mood, anxiety, functioning, and sleep. To aid survey completion, participants received a push notification reminder each evening at 7:00 pm prompting them to open the mindLAMP app and complete the survey. Additionally, we shared weekly data summaries (Fig. 4) with participants, which allowed them to visually track their passive data trends. This approach reinforced engagement by motivating participants to stay active in the study in order to continue observing their own data patterns over time. Engagement materials, including the Python script (Supplementary Materials, “Python Script Implementing Digital Data Reports”) and a sample template of the weekly data summary report (Supplementary Materials, “Template Digital Data Report”) are provided. These tools can be adapted by other research teams and extended to include additional app-based activities such as cognitive games, in-app journaling, or assigned CBT modules.

Notify: continuous real-time data monitoring procedures

To maintain high passive data quality during the study, we applied a real-time monitoring protocol that included: (1) automated data quality report generation and (2) manual dashboard review. This supported early identification and troubleshooting of issues among participants with consistently low passive data quality. Each day, research staff would run a backend Python script that processed GPS data points collected from participants and calculated passive data quality (described in Fig. 1) per participant. Participants were automatically flagged

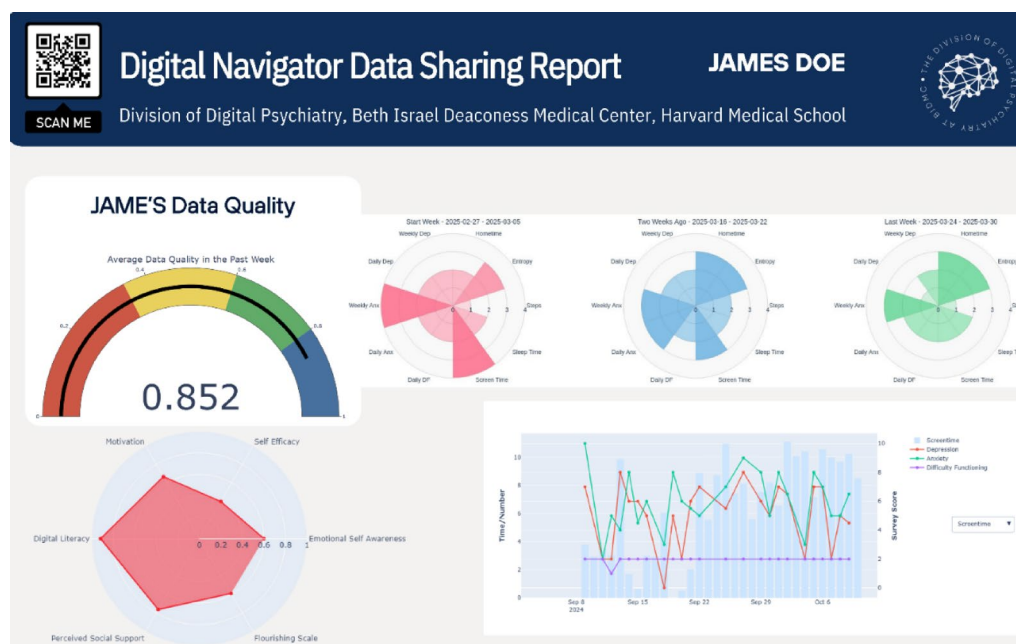


Fig. 4. Sample participant data sharing report used to support engagement in the study. Reports were shared weekly and provided a visual summary of individual passive and active data trends to encourage continued app use and data contribution throughout the study.

for review if passive data quality dropped below 50% (a threshold where feature instability emerges¹¹, remained at 0% for 24 h (indicating complete sensing failure), or no EMA was completed in 3 days (the timeframe after which passive data quality begins to significantly decline without app interaction¹¹). To review participants who recorded low passive data quality, research staff would access the participant's mindLAMP dashboard (Fig. 5) to visually examine surveys completed and passive data trends. This helped the research staff to have a better understanding of the possible source(s) of poor data quality before reaching out to participants with troubleshooting strategies. All monitoring tools used in this process, including the Python scripts used for data quality monitoring and dashboard creation (Supplementary Materials, "Python Scripts Generating Monitoring Dashboard"), and views of the dashboard (Fig. S1, and Fig. S2) are provided online to support replication in other digital phenotyping studies.

Correct: troubleshooting disruptions in passive data collection

Following the identification of disruptions in passive data collection via the mindLAMP dashboard (Fig. 5), research staff initiated a standardized troubleshooting algorithm (Supplementary Materials, "Data Quality Intervention Workflow") to restore data flow and minimize loss. Participants were contacted via email (Supplementary Materials, "Troubleshooting Outreach Email Template") and provided with a custom-developed troubleshooting guide tailored for both Android and iOS devices (Supplementary Materials, "Troubleshooting Checklist"). This guide included step-by-step instructions to assist participants in resolving common technical issues. The troubleshooting protocol focused on three key actions: (1) disabling battery optimization settings or low power mode, (2) verifying that data collection permissions remained enabled, and (3) re-engaging with the app through daily EMA survey completion. While email served as the primary mode of outreach, virtual support via teleconferencing was also available when necessary. To ensure issue resolution and prevent further data loss, follow-up monitoring was conducted after each intervention. This closed-loop process enabled timely detection and correction of passive data collection problems, supporting the integrity and continuity of passive sensing throughout the study. All troubleshooting tools, including email templates, annotated guides, and a decision tree for follow-up actions are available in the Supplementary Materials, which provides a complete and adaptable workflow for research teams to implement.

Statistical procedures

To characterize the passive data quality achieved using the systematized LINC Framework and demonstrate the importance of data quality for reliable feature derivation and valid analytic results, we present our findings in the Results section. All statistical analyses were conducted using R (version 4.4.1) within RStudio (version 2024.04.2). Two-sided p-values were reported throughout, with statistical significance defined as $\alpha < 0.05$ after false discovery rate (FDR) adjustment using the Benjamini-Hochberg procedure.

Descriptive statistics (median, interquartile range [IQR]) were computed to summarize passive data quality, EMA survey completion, and participant demographic and baseline clinical characteristics. Pearson correlation coefficients were used to examine (1) the co-occurrence of GPS and accelerometer data points (Fig. S3) and (2) the relationship between active (survey-based) and passive (GPS-based) data quality (Fig. 7).

To examine how reduced GPS coverage affects the accuracy of derived passive features, high-frequency GPS data from a single participant (1 Hz sampling) were used to generate mobility traces using the GPSmobility R package¹⁶. These data were downsampled to one data point every 30 min to simulate sparser coverage. The resulting mobility traces were visually compared to assess trajectory fragmentation and detection of pause locations. Additionally, to evaluate how sampling frequency impacts home time estimates, the same GPS data were further downsampled at intervals of 5 s, 10 s, 30 s, 1 min, 5 min, 10 min, 15 min, 30 min, 45 min, and 60 min. At each interval, home time was recalculated and compared to the original (1 Hz) estimate. Mean home time error was computed across all available study days.



Fig. 5. Participant-level view of the passive data monitoring dashboard with built-in GPS data heatmap. The left panel displays the most recent timestamps for passive and active data, enabling staff to quickly identify participants with missing or delayed data. The right panel shows a temporal heatmap of GPS data availability (blue means data is present) by time and date, allowing visualization of gaps or patterns in data collection.

To examine how observed passive data quality influences analytic results, we estimated within-person associations between GPS-derived home time and three passive sensing features, screen duration, entropy, and step count, across discrete, non-overlapping bins of passive data quality (0–10%, 10–20%, ..., 90–100%). Associations within each data quality bin were quantified using Spearman rank correlation coefficients to accommodate non-normal feature distributions commonly observed in passive sensing data. To account for repeated daily observations within participants and unequal contributions across bins, uncertainty was estimated using participant-level clustered bootstrap resampling, from which 95% confidence intervals were derived.

Results

Study baseline characteristics

Our analytic cohort for the social media study consisted of 373 participants from an enrolled sample of 417 after 19 participants were disqualified (fraudulent or non-US address) and 25 withdrew from the study. The cohort had a mean age of 21 years ($SD=1.9$) and was predominantly female (277 [74.3%]) and White (198 [53.1%]). Most participants were college students (287 [76.9%]), with 337 (90.3%) owning an iOS smartphone. The sample was non-clinical, with minimal baseline depression (median PHQ-9: 4.0 [IQR 2.0–8.0]) and anxiety (median GAD-7: 4.0 [IQR 2.0–8.0]). See Supplementary Materials (Table S1) for a table of baseline measures.

Study passive data quality

The distribution of participant-level passive data quality over the study period is displayed in Fig. 6. The study achieved a median GPS-based passive data quality of 0.92 (Q2). Of the 373 participants, 79.0% ($n=296$) exceeded 0.50, 75.0% ($n=279$) surpassed 0.59 (Q1), and 25.0% ($n=93$) exceeded 0.98 (Q3). In contrast, 14.2% ($n=53$) had data quality below 0.10, and 9.7% ($n=36$) recorded no GPS-based passive data. Our companion analysis²⁴ reported a median of 0.78 when using all days in the weekly collection periods.

Comparison to prior mindLAMP studies

To contextualize the passive data quality achieved in this study, we compared our results to six prior digital phenotyping studies conducted using the mindLAMP platform (Table 1). These studies varied in sample size ($n=37$ to 695), study duration (4 to 52 weeks), population characteristics (clinical vs. non-clinical), and diagnostic groups. Across these prior studies, median GPS-based passive data quality ranged from 0.12 to 0.80, with most studies reporting medians below 0.60. Our study achieved a median of 0.92, which exceeded all prior mindLAMP studies. Notably, our first quartile ($Q1=0.59$) approached or surpassed the median data quality of most prior studies, indicating that even participants in the lower quartile of our sample achieved data quality comparable to the typical performance in other prior mindLAMP deployments.

Active data quality (survey completion)

Participant engagement with the mindLAMP app was measured by the proportion of daily EMA surveys (active data) completed over the duration of the study. The study achieved a median daily EMA survey completion rate of 0.95 (Q2). Of the 373 participants, 97.0% ($n=361$) had survey completion rates above 0.50, 81.8% ($n=305$) surpassed 0.81 (Q1) and 37.3% ($n=139$) achieved a perfect completion rate of 1.0 (Q3). Only 2.9% ($n=11$) of participants had rates below 0.50, and 1.9% ($n=7$) completed no daily surveys. Our companion analysis²⁴ reported a median of 0.76 when using all days in the weekly collection periods.

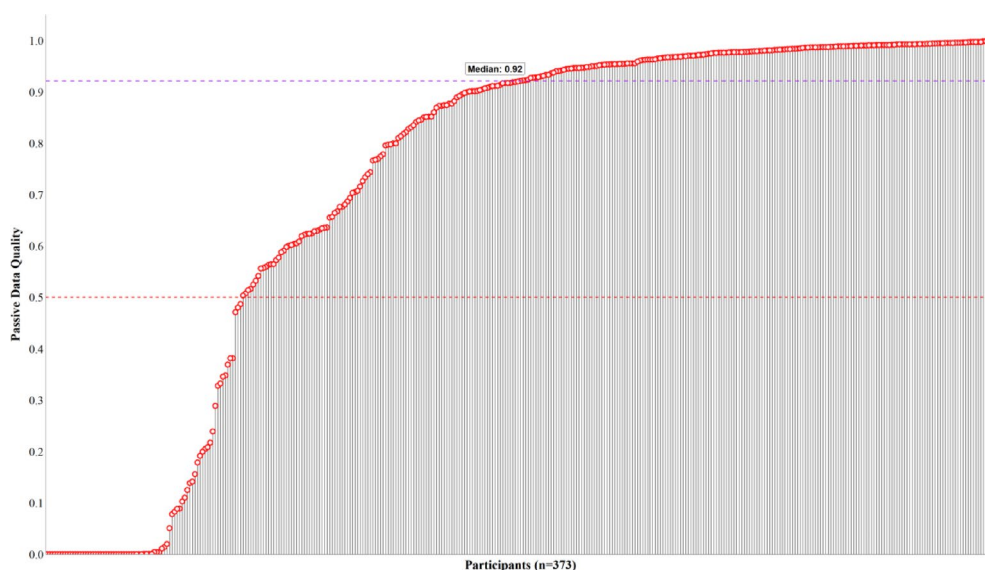


Fig. 6. Distribution of participant-level passive data quality. Each black and orange line segment represents an individual participant's median passive data quality over the study period. The purple dashed line indicates the overall median data quality across all participants (0.92). The red dashed line marks the 0.50 reference point.

Study	Size (n)	Mean Age (yrs)	Date	Length	Sample type	Diagnosis (s)	Sensor type	Median Data quality [†]
Study 1 ²⁹	86	28.9	08/2019 to 05/2022	52 weeks	Clinical	SZ, SZA	GPS, ACC	0.12
Study 2 ³⁰	695	21.5	11/2020 to 05/2021	4 weeks	Non-clinical	None	GPS, ACC	0.43
Study 3 ³¹	96	21.3	11/2021 to 12/2021	4 weeks	Non-clinical	None	GPS, ACC	0.80
Study 4 ³²	76	32.9	08/2021 to 02/2023	52 weeks	Clinical	SSDs	GPS	0.58
Study 5 ³²	70	40.5	08/2022 to 03/2023	8 weeks	Clinical	DEP, ANX	GPS	0.77
Study 6 ³³	37	70.9	04/2022 to 01/2024	4 weeks	Clinical	MCI, DEM	GPS	0.71 ^a
Current study	373	21.0	03/2024 to 03/2025	2/3 weeks	Non-clinical	None	GPS	0.92

Table 1. Passive data quality across mindLAMP digital phenotyping studies. Schizophrenia (SZ); Schizoaffective disorder (SZA); Schizophrenia spectrum disorders (SSDs); Depression (DEP); Anxiety (ANX); Mild cognitive impairment (MCI); Dementia (DEM); Global Positioning System (GPS); Accelerometer (ACC). ^aReported as mean percent of day with GPS data (different metric than median proportion of 10-minute intervals used in other studies). [†]Passive data quality for represented studies, GPS or accelerometer-based.

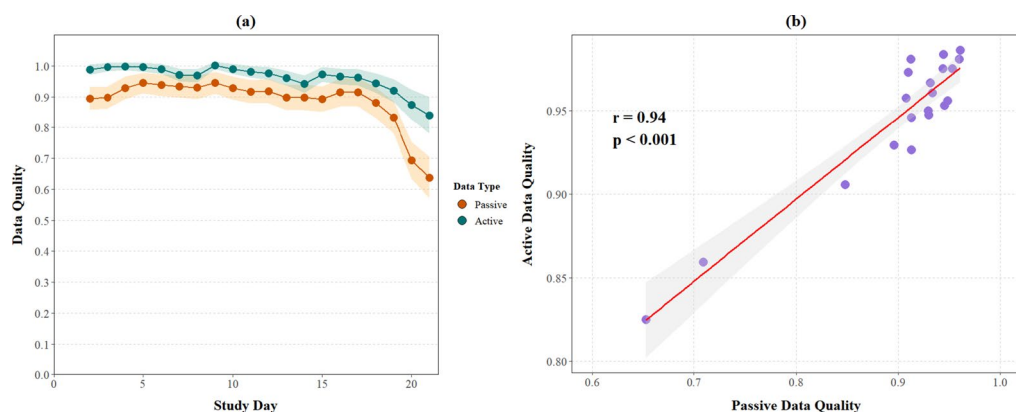


Fig. 7. Relationship between active and passive data quality. (a) Time series of daily data quality for passive (orange) and active (blue) data streams over the 21-day study period. (b) Correlation between median daily passive and active data quality across the study period.

Passive data quality and app engagement

Active engagement with digital phenotyping apps is essential for maintaining passive data collection. Modern smartphone operating systems implement battery optimization policies that restrict background processes including GPS updates, accelerometer access, and sensor data transmission for apps not regularly used. By requiring daily app interaction through EMA completion, the Interact component of LINC maintains the app in active status, preventing operating system suspension of background data collection. Figure 7 displays the temporal relationship between daily active and passive data quality over the study period. Day 1, representing partial enrollment during device configuration, showed reduced quality for both metrics (passive: 0.32, active: 0.89) and was excluded from visualizations to focus on complete study days (2–21).

Following Day 1, both active engagement and passive data collection rapidly achieved high data quality, with both exceeding 0.90 by Day 2. This early attainment of high data quality underscores the role of LINC's Launch component in supporting high-quality data collection during the initial study period, emphasizing standardized onboarding and comprehensive device and app configuration. Throughout the active study period (Days 2–21), active data quality remained consistently high alongside passive data quality. This strong positive correlation between active and passive data quality (Spearman's $r = 0.94$, 95% CI: 0.85–0.98, $p < 0.001$) demonstrates the interdependence between app engagement and sustained background data collection. Both declined during the final study days (Days 18–21), consistent with reduced participant engagement as the study conclusion approached. This natural temporal relationship supports the framework's emphasis on sustained app engagement with EMAs to maintain high-quality passive sensing.

Monitoring and intervention

Beyond participant engagement, device-specific settings and behaviors such as entering Low Power Mode or disabling sensor permissions can disrupt passive data collection. To identify and address these technical issues, research assistants systematically monitored passive data quality using automated scripts and dashboards and contacted participants who had persistently low quality (below 0.50, remained at 0% for 24 h, or no EMA completion for 3 days) to troubleshoot potential causes. Of the 373 participants, 31.6% ($n = 118$) required no intervention, and an additional 35.7% ($n = 133$) were contacted only once, indicating that two-thirds of

participants (67.3%) needed minimal to no troubleshooting support. An additional 17.2% ($n=64$) required two contacts, while 15.5% ($n=58$) needed three or more contacts. The mean number of troubleshooting contacts per participant was 1.3 (SD=1.2) across the entire cohort. Over the study period, which ran for approximately one year, a total of 468 interventions were conducted (approximately 9 per week), demonstrating the modest operational burden of the monitoring protocol. The observed contact frequency and weekly intervention load suggest that systematic monitoring approaches of this nature may be implementable without imposing substantial burden on either participants or research staff.

Of the 255 participants who were contacted at least once during the study, 179 (48.0% of total sample) responded positively to troubleshooting interventions and subsequently maintained data quality above 0.50. Among these 179 participants, the most common resolvable issue encountered was Low Power Mode being inadvertently enabled. However, 40 participants (10.7% of total sample) continued to experience passive data quality below 0.50 despite intervention attempts, predominantly due to outdated smartphone models with limited sensor capabilities or restrictive operating system versions. Additionally, 36 participants (9.7% of total sample) collected no passive data throughout the study. These cases of complete missingness were expected given the “Bring Your Own Device” (BYOD) nature of the study and primarily reflected device incompatibility with the mindLAMP app or, to a lesser extent, sustained non-response to repeated contact attempts.

Figure 8. Illustrative effects of the first research staff outreach attempt on passive data quality. This figure presents selected examples to illustrate the range of intervention scenarios and outcomes observed in the study and is not intended to quantify intervention effects. Shaded regions indicate the timing of research assistant outreach, categorized as early (top row, teal), late (middle row, purple), or futile (bottom row, yellow). Participants A–C (top row) illustrate early outreach followed by recovery of data quality; participants D–F (middle row) show delayed outreach with variable recovery; and participants G–I (bottom row) show persistently low or absent data quality despite outreach attempts, consistent with expected non-remediable cases.

Passive data quality and derived features

To illustrate the importance of achieving high passive data quality in digital phenotyping studies that aim to make inferences about behavior, we conducted a controlled downsampling analysis to examine the sensitivity of GPS-derived features to progressively lower levels of data quality. This case-based methodological approach

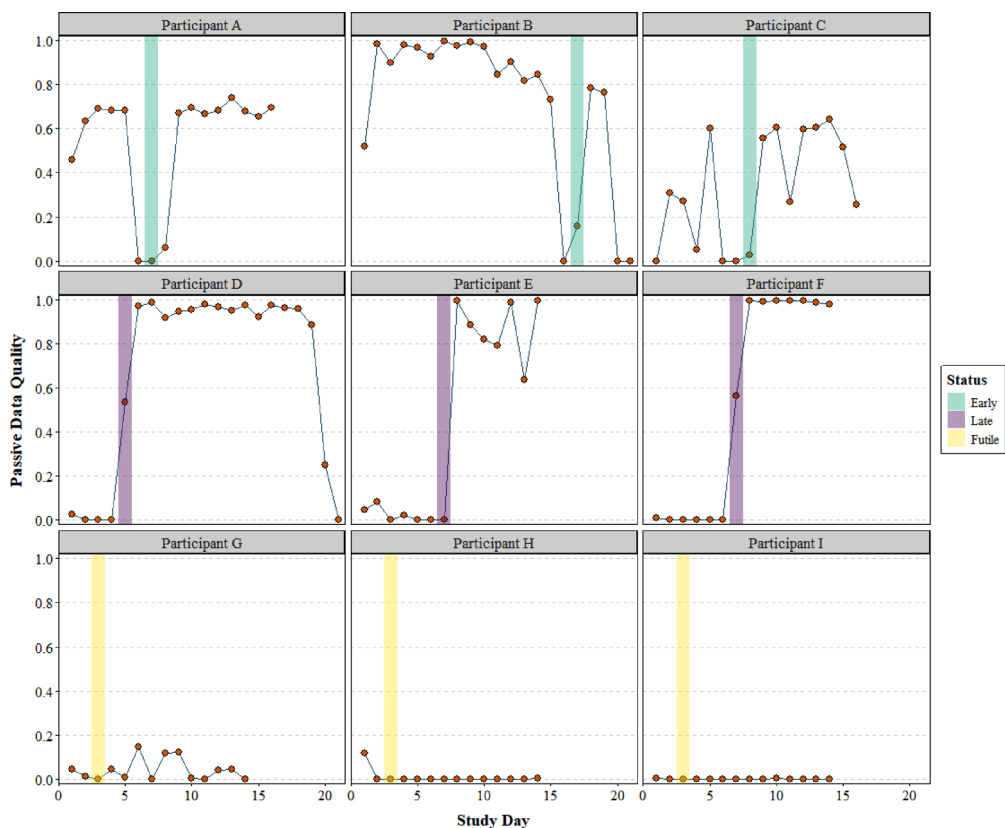


Fig. 8. presents nine representative examples selected to illustrate the diversity of intervention scenarios and outcomes. Participants A–C (top row) demonstrate early intervention followed by variable responses in passive data quality. Participants D–F (middle row) illustrate delayed responses to sustained low passive data quality. Participants G–I (bottom row) depict cases of persistently zero data quality despite intervention attempts, reflecting expected outcomes in participants with device incompatibility and representative of the 36 individuals who contributed no passive data.

isolates the effect of reduced data quality on feature reliability while holding all other factors constant, including the participant, study day, and underlying behavioral patterns. We selected GPS data from a participant with a median data quality above 0.90 (representative of the study's third quartile) and focused on a day with 0.96 data quality (1 Hz sampling, 96% of 10-minute intervals with valid GPS data). Using this high-quality baseline, we systematically downsampled the GPS data at intervals ranging from 5 s to 60 min, simulating progressively lower data quality levels. For each downsampling interval, we calculated home time estimates and compared them to the ground truth derived from the original 1 Hz data. Figure 9 illustrates the relationship between sampling interval and home time estimation error. At high-frequency sampling (5–10 s), home time error remained near zero with minimal variability, indicating that estimates were highly accurate and stable. However, as sampling intervals lengthened beyond 10 min, both median error and variability increased substantially. At 30–60 min intervals (equivalent to data quality of approximately 0.50 or lower), home time was consistently underestimated by several hours, with high variability across estimates. This demonstrates that temporal gaps in GPS sampling, characteristic of poor data quality, substantially degrade the accuracy of derived behavioral features such as home time.

To visualize the mechanism underlying this degradation, Fig. 10 presents mobility plots comparing the same 24-hour period under two sampling conditions. Panel (a) shows the original 1 Hz sampling (data quality of 0.96), revealing complete movement trajectories with clear identification of sustained locations: probable home location (dark blue circle, 8 PM–6 AM) and probable work location (large red circle, 9 AM–4 PM). Panel (b) shows the identical day downsampled to 30-minute intervals, the threshold at which home time estimates became unreliable in Fig. 9. This sparse sampling resulted in visibly fragmented trajectories, reduced detection of stationary periods during the workday (smaller, more diffuse red circles), and less reliable identification of home location (scattered blue circles). Of note, this systematic downsampling represents an idealized form of missingness with regular intervals. Real-world data quality below 0.50 typically exhibits irregular, unpredictable gaps that may introduce additional artifacts beyond those shown here. Thus, this analysis demonstrates that the high data quality achieved in this study (median 0.92, with 75% exceeding 0.59) enables reliable extraction of behavioral features, while quality below 0.50 introduces substantial uncertainty in derived measures such as home time and mobility entropy.

Impact of data quality on feature-based analyses

Passive data quality affects not only individual feature accuracy but also the reliability of relationships between features. To evaluate this, we examined correlations between GPS-derived home time and three passive sensing features (screen duration, mobility entropy, and step count) across participants grouped by observed GPS data quality (0–10%, 10–20%, ..., 90–100% bins, based on actual achieved data quality with no downsampling). Here, Fig. 11 illustrates the sensitivity of feature-based analyses to empirically observed passive data quality. Across all three GPS-derived features, associations with home time varied systematically as data quality decreased. At higher data quality levels, correlations were directionally consistent and relatively stable, with narrower confidence intervals. In contrast, at lower data quality levels, particularly below approximately 50% passive data quality, associations were attenuated and increasingly unstable, accompanied by wider confidence intervals. Mobility entropy showed a progressively stronger and more stable negative association with home time as data quality

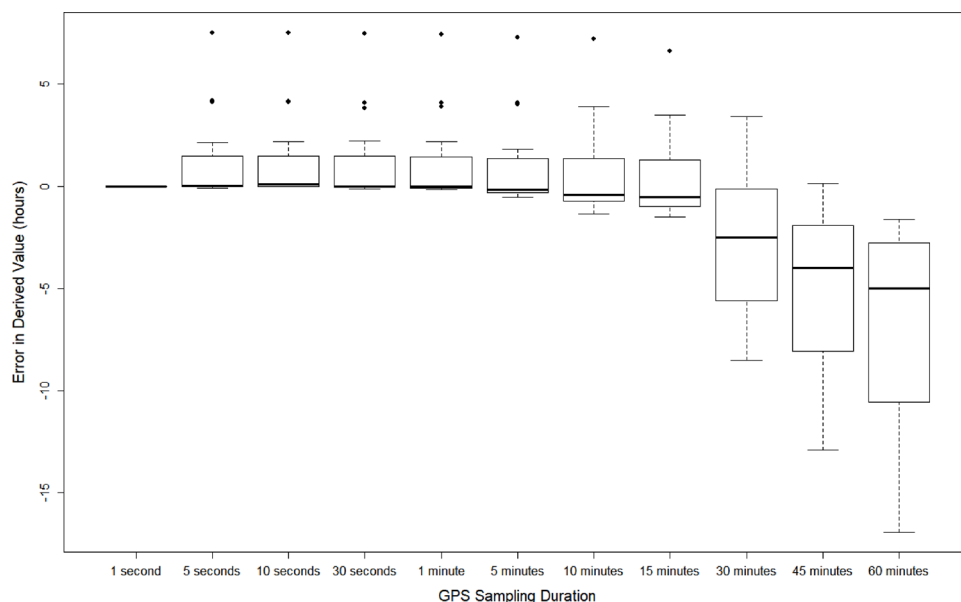


Fig. 9. Relationship between GPS sampling duration and error in derived home time estimates. GPS data from a single participant day were progressively downsampled at intervals ranging from 1 s to 60 min. Each boxplot displays the distribution of error in home time (in hours) introduced by downsampling at each interval, relative to the original high-resolution (1 Hz) dataset.

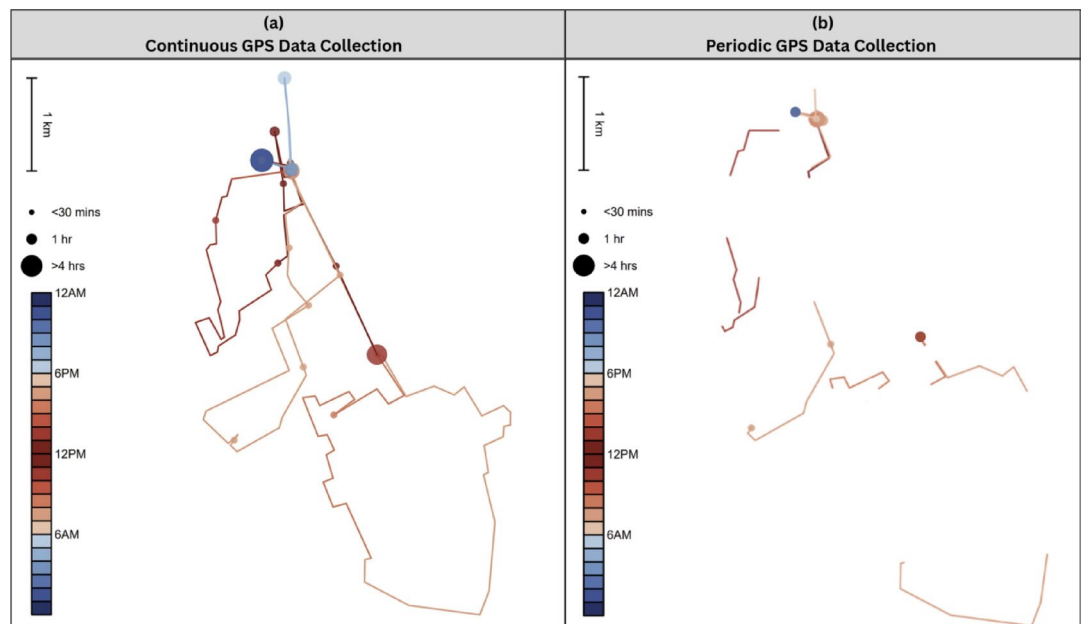


Fig. 10. Mobility trajectories generated from GPS data collected using two sampling strategies on the same participant day. **(a)** Continuous GPS data collection at 1 Hz. **(b)** Periodic GPS data collection at a rate of one point every 30 min. Lines represent movement between logged GPS coordinates. Circles represent detected pause locations, with circle size indicating estimated duration of the pause and circle color representing the time of day. Arrows represent points of fragmentation in mobility trajectory due to missing GPS data points. The color scale ranges from 12 AM (dark blue) to 11:59 PM (dark red). To protect participant privacy and prevent potential reidentification through full trajectory shape or missingness patterns, mobility traces were manually distorted and are therefore presented for illustrative purposes only.

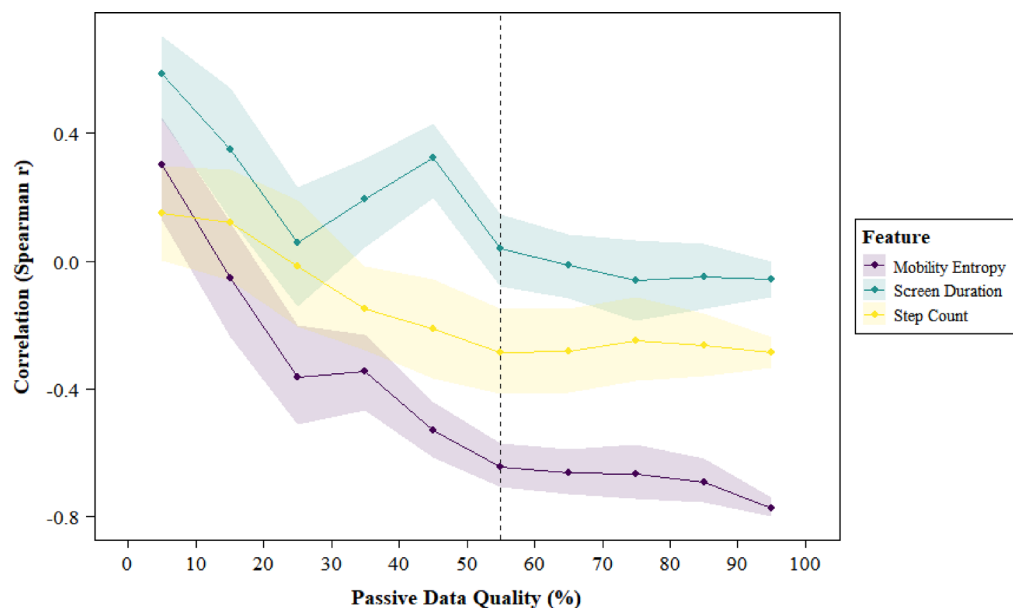


Fig. 11. Sensitivity of feature-based analyses to passive data quality. Within-person Spearman correlations between GPS-derived home time and three passive sensing features (mobility entropy, screen duration, and step count) are shown across discrete bins of empirically observed passive data quality. Points indicate correlation estimates within each data quality bin, and shaded regions represent participant-clustered 95% confidence intervals accounting for repeated daily observations. Associations are relatively stable at higher data quality levels but become attenuated and increasingly uncertain at lower data quality, particularly below approximately 50–55% passive data quality (vertical dashed line), indicating reduced reliability of feature-based inference under low data quality conditions.

improved, whereas at lower data quality levels the association was weaker and more variable. Step count showed a similar pattern of attenuation and reduced stability at lower data quality, while screen duration demonstrated comparatively smaller but more stable associations at higher data quality. Together, these patterns indicate that reduced passive data quality is associated with increased variability and decreased reliability of feature-based analytic results, rather than reflecting consistent differences in the underlying behavioral relationships that exist.

Discussion

In this paper, we introduced LINC (Launch, Interact, Notify, and Correct), a framework of systematized best practices paired with practical tools for monitoring and supporting passive data collection in smartphone-based digital phenotyping studies. Applied to an observational study examining social media use and youth mental health ($n=373$), this structured approach achieved a median GPS-based passive data quality of 0.92 (IQR: 0.59–0.98), with 75% of participants exceeding 0.59 (Q1), and 25% surpassing 0.98 (Q3). Our findings not only demonstrate the feasibility of implementing passive data quality management frameworks but also provide insights into data quality thresholds that meaningfully affect the reliability of feature-based analyses.

Comparison to prior digital phenotyping studies

The social media study achieved a median data quality of 0.92, which compares favorably to most prior mindLAMP digital phenotyping studies (Table 1), which report median GPS-based passive data quality ranging from 0.12 to 0.80 across clinical and non-clinical populations. Notably, our first quartile (Q1 = 0.59) approached or exceeded the median quality of most prior studies, indicating that even participants in the lower quartile achieved data quality comparable to typical performance in other mindLAMP deployments. However, while promising, multiple factors beyond the framework may have contributed to this difference, including study duration (2–3 weeks versus up to 52 weeks in prior studies), population characteristics (non-clinical young adults versus clinical samples), recruitment methods, and temporal improvements in the mindLAMP platform itself. Thus, causal attribution to LINC components requires controlled comparison against standard practices.

Feasibility evidence for the LINC framework

Temporal patterns observed in the data were consistent with the conceptual foundations of each LINC component, though causal conclusions cannot be drawn without controlled comparisons. The “Launch” component emphasized standardized device configuration and permission settings at enrollment. Consistent with this approach, both active and passive data quality exceeded 90% by Day 2 following initial onboarding. This early attainment of high data quality may reflect the role of proper initial configuration in supporting immediate passive data collection, although other contributing factors, such as participant motivation or device characteristics, cannot be ruled out.

The “Interact” component is based on the observation that modern smartphone operating systems employ aggressive battery optimization strategies that can restrict background processes for infrequently used applications. By incorporating daily EMA completion, this component is intended to promote regular app engagement, which may help reduce operating system suspension of passive sensing processes. The strong positive correlation between active engagement and passive data quality observed across the study period (Spearman's $r=0.94$) is consistent with this interpretation, as is the parallel decline in both metrics during Days 18–21 as participants approached the scheduled study endpoint. However, this association may also reflect unmeasured participant-level factors, such as individual differences in conscientiousness or device characteristics, that could influence both engagement and passive data quality concurrently.

The “Notify” and “Correct” components involve real-time monitoring with predefined quality thresholds (below 0.50, at 0% for 24 h, or no EMA completion for 3 days) and systematic troubleshooting when participants are flagged. Majority of participants contacted (70.2%) had a positive outcome, and were able to maintain data quality above 0.50 post intervention, while 29.8% either remained below 0.50 or collected no data. Majority of participants (77.3%) were only contacted once or twice, with the most common issue only requiring participants to charge their phones or to disable Low Power Mode. The modest intervention burden of 1.3 contacts per participant (approximately nine per week) suggests that implementing the LINC framework may be operationally feasible, as flagged participant follow-ups were managed by only two research assistants on a weekly basis. Although approximately 30% of contacts did not result in sustained resolution of data collection issues, these instances largely reflect inherent limitations of BYOD approaches, such as outdated or incompatible smartphones, which structured frameworks may help identify but cannot fully address.

Methodological insights for digital phenotyping research

High-quality passive data is essential for supporting accurate feature derivation and interpretable downstream analyses in digital phenotyping research. In our downsampling analyses, GPS-derived features become increasingly inaccurate at higher levels of missingness. In particular, estimates of home time became substantially less accurate when sampling intervals exceeded 30 min (50% data quality), underestimating values by two to four hours. In contrast, sampling intervals of 15 min or less were associated with comparatively accurate estimates. These observations are consistent with prior work by Currey and Torous¹¹, who reported that greater missingness in GPS data was associated with unreliable estimates of home time and entropy, and together provide empirical guidance for minimum passive data quality thresholds in studies using GPS-derived behavioral features.

Complementary correlation analyses across empirically observed levels of passive data quality further indicated that associations between GPS-derived home time and other passive features, including screen duration, entropy, and step count, became more stable above 50% passive data quality. Below this threshold, correlation estimates were generally attenuated and showed increased variability. This pattern aligns with findings from Shen and colleagues¹⁵, who observed stabilization of regression coefficients involving home time

and cadence under conditions of higher data quality (i.e., lower missingness) in models predicting functional status. Thus, these results indicate that passive data quality may be closely related not only to the accuracy of individual feature estimates but also to the stability and interpretability of multi-feature analyses and the reproducibility of findings in digital phenotyping research.

Limitations and future directions

The implementation of the LINC Framework had several limitations worth mentioning. First, the absence of a concurrent control group receiving standard onboarding without the systematized LINC approach limits causal inference. Although the observed levels of passive data quality exceed those reported in many prior mindLAMP studies, these differences cannot be definitively attributed to the LINC components as opposed to participant characteristics, study duration, platform updates, or other unmeasured factors. Therefore, randomized controlled trials comparing LINC against standard practices are needed to establish causal effects and quantify the framework-attributable improvements.

Second, generalizability is limited by the characteristics of our study sample. Participants were young adults aged 18 to 24, primarily college students, with minimal baseline psychiatric symptoms, reflecting a non-clinical population. Additionally, the sample was predominantly iOS users, which may have contributed to more consistent passive data collection given differences in how Android and iOS handle background app processes. Patterns of passive data quality, responsiveness to monitoring and follow-up, and the practical utility of the framework may differ in clinical populations, older adults, individuals with lower levels of technological literacy, or samples with greater Android representation^{26,33}. In particular, the relatively high baseline app engagement observed in this sample may have supported the Interact component, but similar engagement levels may be less likely in populations with greater symptom burden, lower motivation, or competing demands on attention and time. Studies should examine framework performance in clinical samples (e.g., individuals with schizophrenia, depression, anxiety disorders), older adults, populations with varying technological literacy, and international contexts with different device ecosystems and cultural norms around smartphone use.

Third, the relatively short study duration (2–3 weeks) limits assessment of whether approaches like LINC would be feasible over extended periods. Participant engagement and passive data quality may naturally decline in longer studies regardless of monitoring efforts, and the burden of sustained troubleshooting may become untenable as study duration extends. Validation in studies of varying durations (weeks to months to years) is needed to establish long-term viability. This would help determine whether frameworks like LINC remain effective as initial enrollment novelty wears off and whether intervention burden scales sustainably over time.

Finally, the LINC framework was developed and tested specifically on the mindLAMP platform. While general principles (standardized configuration, engagement requirements, real-time monitoring, systematic troubleshooting) are broadly applicable, specific implementation details will likely require adaptation for other digital phenotyping platforms with different technical architectures, data collection mechanisms, or sensor access protocols. Thus, evaluation of how LINC translates to other digital phenotyping platforms (e.g., Beibe¹, AWARE³⁴, RADAR-MDD³⁵) would establish broader applicability and identify platform-specific adaptations required for effective implementation³⁶.

Conclusion

In closing, the LINC framework provides a structured, operational approach for monitoring and supporting passive data collection in smartphone-based digital phenotyping studies. By systematizing commonly used but inconsistently reported practices into a unified workflow with practical implementation resources, LINC offers a feasible strategy for improving data continuity in real-world research settings. While causal effects cannot be inferred in the absence of controlled comparisons, our findings illustrate how proactive data quality management may support more reliable feature derivation and downstream analyses. Importantly, LINC is intended to complement rather than replace post hoc imputation and missing data methods by reducing extreme and systematic missingness before analytic modeling. Future work evaluating LINC across diverse populations, longer study durations, and alternative digital phenotyping platforms will be essential to establish broader applicability and inform best practices for passive data collection.

Data availability

An anonymized and de-identified dataset without personal health information can be shared by the corresponding author upon reasonable request, subject to IRB approval and a data use agreement.

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Author contributions

EC : formal analysis, visualization, writing original draft, and writing, review & editing. EL : writing original draft, and writing, review & editing. MF : software, data curation, writing, review & editing. JT : conceptualization, funding acquisition, methodology, project administration, resources, supervision, validation, writing original draft, and writing, review & editing.

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Declarations

Competing interests

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Additional information

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